

Respiratory Health and Measurement Engineering

Airway, lung, gas-exchange, pulmonary vascular, sleep-breathing, and ventilatory-pump systems

Volume 03B of the Health Atlas Research Suite

Purpose

Provide a causal, clinical, and engineering map of respiratory health that connects breathing mechanics, gas exchange, host defense, pulmonary circulation, sleep and exercise reserve to validated measurement and safe device development.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This synthesis uses current major respiratory guidance and measurement standards through 13 August 2026. It does not reproduce disease-specific guidelines, dosing tables, ventilator protocols or local emergency pathways. Device sections describe research and engineering requirements, not authorization for unsupervised human use or diagnosis.

How to use this volume

Respiratory health is not equivalent to a normal oxygen saturation or a single spirometry value. It is the capacity to move air, exchange gases, protect the airway, adapt ventilation to metabolic demand, sustain sleep and activity, survive exposures and infection, and recover without excessive work or long-term loss of reserve. A complete profile therefore combines symptoms and function with airflow, lung volumes, diffusion, oxygen and carbon-dioxide physiology, respiratory muscle capacity, imaging, inflammation or microbiology when indicated, sleep and exercise challenge, environment, trajectory and comorbidity. This volume treats every device output as an inference whose validity depends on calibration, physiology, artifact, intended use and reference method.

Questions this volume is designed to answer

1. What dimensions constitute respiratory health beyond oxygen saturation?
2. How do airway resistance, compliance, ventilation, perfusion, diffusion, respiratory muscles and neural control interact?
3. Which determinants shape peak lung function, decline, exacerbation, recovery and inequity?
4. How should common respiratory syndromes be connected to mechanism rather than treated as isolated labels?
5. What does each respiratory device physically sense, and what can it not establish?
6. How should a safe multimodal respiratory platform be built, calibrated and validated?
7. Which interventions improve survival, function or symptoms, and where are precision claims still uncertain?

Scope

- Normal respiratory anatomy, mechanics, gas exchange, pulmonary circulation, defense and control.
- Lung development, peak function, aging, reserve, exercise, sleep and adaptation to altitude/environment.
- Obstructive, restrictive, infectious, vascular, pleural, neoplastic, sleep-related, neuromuscular and post-acute phenotypes.
- Clinical assessment and longitudinal profiling from symptoms and examination through imaging and invasive reference methods.
- Device physics, signal chains, calibration and validation for airflow, pressure, gas, optical, acoustic, motion and biochemical sensing.
- Prevention, pharmacology, oxygen/ventilation, rehabilitation, procedures, transplantation and palliative care.
- Research architecture for multimodal home and laboratory monitoring.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.

- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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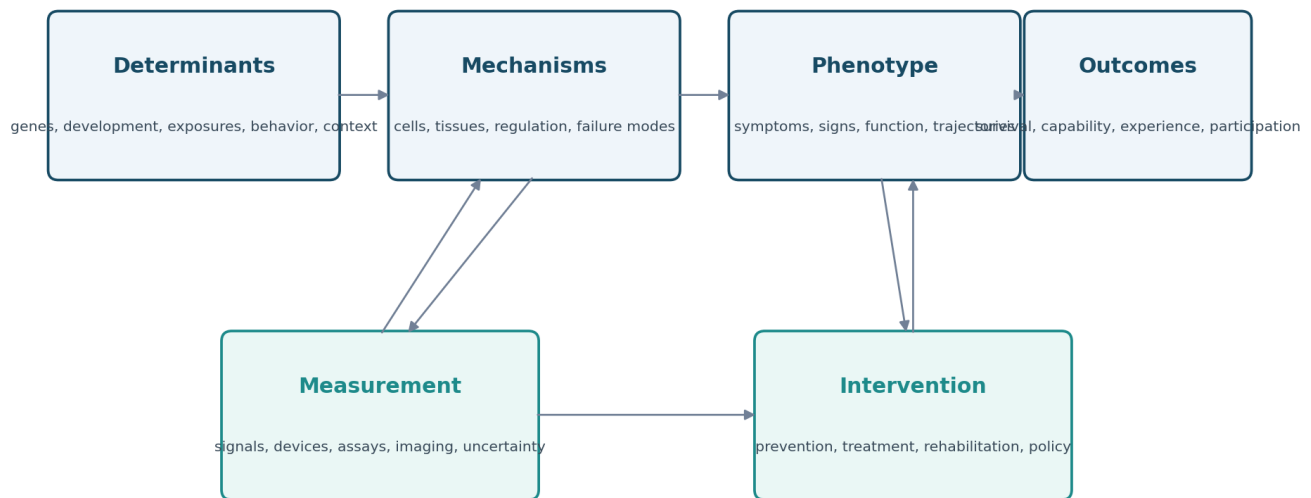
Executive synthesis

Respiratory health is not equivalent to a normal oxygen saturation or a single spirometry value. It is the capacity to move air, exchange gases, protect the airway, adapt ventilation to metabolic demand, sustain sleep and activity, survive exposures and infection, and recover without excessive work or long-term loss of reserve. A complete profile therefore combines symptoms and function with airflow, lung volumes, diffusion, oxygen and carbon-dioxide physiology, respiratory muscle capacity, imaging, inflammation or microbiology when indicated, sleep and exercise challenge, environment, trajectory and comorbidity. This volume treats every device output as an inference whose validity depends on calibration, physiology, artifact, intended use and reference method.

Central organizing principle

Provide a causal, clinical, and engineering map of respiratory health that connects breathing mechanics, gas exchange, host defense, pulmonary circulation, sleep and exercise reserve to validated measurement and safe device development.

Respiratory Health and Measurement Engineering



A health profile is a causal, temporal, functional, and measurement model - not a single score.

Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.

- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.
- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Severe breathing difficulty, altered mental status, cyanosis, apnea, stridor, suspected tension pneumothorax or pulmonary embolism, rapidly falling oxygenation, or ventilatory failure are emergencies. Consumer sensors must not delay clinical escalation. Human-connected respiratory devices require appropriate isolation, cleaning, alarm, usability and risk controls.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Antiquity to 1600s	Breath, pulse, chest motion, and sputum interpreted through humoral and vital theories.	Established respiratory symptoms as observable clinical phenomena and linked breathing to life.	Mechanisms of ventilation, circulation, infection, and gas exchange were conflated; anatomy was limited.
1600s-1700s	Harveyan circulation, Boyle's gas law, Hooke and Lower experiments, and early microscopy.	Created a mechanical and experimental model of lung inflation and blood oxygenation.	Gas composition, diffusion, cellular metabolism, and pulmonary microanatomy remained incomplete.
1816 onward	Laennec developed the stethoscope and correlated sounds with pathology.	Enabled noninvasive localization of airway, pleural, and parenchymal disease.	Auscultation is observer-dependent, nonspecific, and relatively insensitive for early disease.
1800s	Spirometry, gas analysis, pathology, germ theory, and radiography emerged.	Made lung volumes, airflow, infection, tuberculosis, and structural lesions measurable.	Reference values were narrow and exposure, equity, and longitudinal reserve were poorly represented.
1900-1950	Oxygen therapy, bronchoscopy, thoracic surgery, mechanical ventilation, antibiotics, and arterial blood-gas analysis.	Enabled rescue of hypoxemia and direct airway assessment.	Iatrogenic lung injury, resistance, and chronic disease burdens became apparent.
1950-2000	Body plethysmography, diffusion testing, CT, polysomnography, pulmonary rehabilitation, inhaled therapies, and intensive-care ventilation matured.	Separated obstruction, restriction, diffusion, sleep-related breathing disorders, and ventilation failure.	Disease labels often remained siloed despite shared exposures and multimorbidity.
2000-2020	Molecular phenotyping, high-resolution CT, portable oximetry, home sleep testing, biologics, and lung-protective ventilation.	Supported endotypes, earlier structural assessment, remote monitoring, and targeted therapy.	Biomarker-defined response does not guarantee broad population benefit or equitable access.
2020-2026	COVID-19, long-term respiratory sequelae, rapid diagnostics, wastewater surveillance, telehealth, and renewed focus on indoor air.	Revealed coupling among infection, immunity, thrombosis, ventilation, environment, and social structure.	Post-acute syndromes, causal subtypes, and optimal monitoring remain incompletely resolved.

2.0.1 Breath, pulse, chest motion, and sputum interpreted through humoral and vital theories.

Context. Observation preceded reliable instrumentation.

Scientific consequence. Established respiratory symptoms as observable clinical phenomena and linked breathing to life.

What remained unresolved. Mechanisms of ventilation, circulation, infection, and gas exchange were conflated; anatomy was limited.

Evidence anchors: [3,5]

2.0.2 Harveyan circulation, Boyle's gas law, Hooke and Lower experiments, and early microscopy.

Context. Quantitative physics began replacing vital explanations.

Scientific consequence. Created a mechanical and experimental model of lung inflation and blood oxygenation.

What remained unresolved. Gas composition, diffusion, cellular metabolism, and pulmonary microanatomy remained incomplete.

Evidence anchors: [3,4]

2.0.3 Laennec developed the stethoscope and correlated sounds with pathology.

Context. The instrument transformed the body into an acoustic diagnostic object.

Scientific consequence. Enabled noninvasive localization of airway, pleural, and parenchymal disease.

What remained unresolved. Auscultation is observer-dependent, nonspecific, and relatively insensitive for early disease.

Evidence anchors: [4,5]

2.0.4 Spirometry, gas analysis, pathology, germ theory, and radiography emerged.

Context. Measurement and public-health control developed together.

Scientific consequence. Made lung volumes, airflow, infection, tuberculosis, and structural lesions measurable.

What remained unresolved. Reference values were narrow and exposure, equity, and longitudinal reserve were poorly represented.

Evidence anchors: [5,8]

2.0.5 Oxygen therapy, bronchoscopy, thoracic surgery, mechanical ventilation, antibiotics, and arterial blood-gas analysis.

Context. Acute care capabilities expanded faster than long-term systems thinking.

Scientific consequence. Enabled rescue of hypoxemia and direct airway assessment.

What remained unresolved. Iatrogenic lung injury, resistance, and chronic disease burdens became apparent.

Evidence anchors: [5,12]

2.0.6 Body plethysmography, diffusion testing, CT, polysomnography, pulmonary rehabilitation, inhaled therapies, and intensive-care ventilation matured.

Context. Technical specialization produced powerful but fragmented measurement chains.

Scientific consequence. Separated obstruction, restriction, diffusion, sleep-related breathing disorders, and ventilation failure.

What remained unresolved. Disease labels often remained siloed despite shared exposures and multi-morbidity.

Evidence anchors: [6,7,8,9]

2.0.7 Molecular phenotyping, high-resolution CT, portable oximetry, home sleep testing, biologics, and lung-protective ventilation.

Context. Precision and digital approaches entered routine respiratory care.

Scientific consequence. Supported endotypes, earlier structural assessment, remote monitoring, and targeted therapy.

What remained unresolved. Biomarker-defined response does not guarantee broad population benefit or equitable access.

Evidence anchors: [6,7,24]

2.0.8 COVID-19, long-term respiratory sequelae, rapid diagnostics, wastewater surveillance, telehealth, and renewed focus on indoor air.

Context. The pandemic made respiratory health a systems and infrastructure problem.

Scientific consequence. Revealed coupling among infection, immunity, thrombosis, ventilation, environment, and social structure.

What remained unresolved. Post-acute syndromes, causal subtypes, and optimal monitoring remain incompletely resolved.

Evidence anchors: [10,12,15]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Conducting-airway mechanics

Core principle. Airway caliber, branching geometry, surface liquid, smooth muscle, and dynamic compression govern resistance and flow distribution.

Control architecture. Autonomic tone, inflammatory mediators, epithelial signals, lung volume, surfactant, and respiratory muscle pressure regulate caliber.

Failure modes. Bronchoconstriction, mucus, edema, wall remodeling, foreign body, collapse, or intrathoracic pressure can sharply raise resistance.

Observable consequences. Flow-volume loop shape, wheeze, oscillometry, airway resistance, regional ventilation, work of breathing.

Cross-system coupling. Allergy, infection, reflux, autonomic state, sleep, exercise, and cardiac filling alter airway behavior.

Evidence anchors: [3,4,8]

3.0.2 Alveolar ventilation

Core principle. Minute ventilation is partitioned into effective alveolar ventilation and dead-space ventilation; carbon dioxide tracks the balance between production and alveolar clearance.

Control architecture. Central and peripheral chemoreflexes, mechanoreceptors, behavioral drive, sleep state, and respiratory muscles coordinate ventilation.

Failure modes. Hypoventilation, excessive dead space, muscle weakness, central depression, or obstruction raise arterial carbon dioxide.

Observable consequences. Respiratory rate, tidal volume, capnography, arterial or venous gases, ventilatory response testing.

Cross-system coupling. Metabolic rate, acid-base status, drugs, neuromuscular function, obesity, and altitude modify demand.

Evidence anchors: [3,4]

3.0.3 Gas exchange and diffusion

Core principle. Oxygen and carbon dioxide move across the alveolar-capillary membrane according to gradients, surface area, thickness, solubility, and capillary transit.

Control architecture. Ventilation-perfusion matching, hypoxic pulmonary vasoconstriction, capillary recruitment, hemoglobin, and cardiac output preserve transfer.

Failure modes. Shunt, diffusion limitation, low inspired oxygen, ventilation-perfusion mismatch, anemia, or dyshemoglobinemia impair oxygen delivery.

Observable consequences. Pulse oximetry, blood gas, diffusing capacity, alveolar-arterial gradient, exercise desaturation, imaging.

Cross-system coupling. Hematology, cardiovascular output, mitochondrial demand, altitude, and toxic exposure determine clinical consequence.

Evidence anchors: [3,4,8]

3.0.4 Pulmonary circulation

Core principle. A low-pressure, high-compliance vascular bed accepts the entire cardiac output and recruits with exercise.

Control architecture. Endothelium, nitric oxide, oxygen tension, left-sided filling pressure, lung volume, and thrombosis regulate pressure and resistance.

Failure modes. Vascular obstruction, remodeling, hypoxia, left-heart disease, or parenchymal loss produce pulmonary hypertension and right-heart stress.

Observable consequences. Echocardiographic estimates, right-heart catheterization, CT angiography, exercise response, biomarkers.

Cross-system coupling. Cardiovascular, thrombotic, autoimmune, hepatic, sleep, and altitude processes converge here.

Evidence anchors: [2,3,4]

3.0.5 Respiratory muscles and pump

Core principle. Diaphragm, intercostal, accessory, upper-airway, and abdominal muscles generate pressure against elastic and resistive loads.

Control architecture. Motor neurons, neuromuscular junction, energy supply, chest-wall geometry, and feedback control sustain reserve.

Failure modes. Weakness, fatigue, pain, deformity, obesity, central drive failure, or load-capacity imbalance causes ventilatory failure.

Observable consequences. Maximal inspiratory/expiratory pressure, sniff pressure, ultrasound, electromyography, spirometry, transdiaphragmatic pressure.

Cross-system coupling. Neurologic, muscular, nutritional, metabolic, and sleep disorders commonly present through the pump.

Evidence anchors: [3,8]

3.0.6 Elastic recoil and compliance

Core principle. Lung tissue, surfactant, chest wall, and pleural pressure determine pressure-volume behavior and resting lung volume.

Control architecture. Alveolar type II cells, extracellular matrix turnover, posture, muscle tone, and developmental geometry regulate compliance.

Failure modes. Fibrosis decreases compliance; emphysema increases lung compliance while reducing recoil; edema and atelectasis alter both.

Observable consequences. Static/dynamic compliance, pressure-volume curves, lung volumes, imaging, crackles, ventilator mechanics.

Cross-system coupling. Connective tissue, aging, prematurity, fluid balance, and mechanical ventilation shape elastic behavior.

Evidence anchors: [3,5,41]

3.0.7 Mucociliary and epithelial defense

Core principle. The epithelium is a physical, chemical, sensory, and immune interface; cilia and mucus clear particles and microbes.

Control architecture. Hydration, ion transport, ciliary beat, cough, innate immunity, microbiota, and epithelial repair maintain clearance.

Failure modes. Smoking, dehydration, genetic ion-channel defects, toxins, infection, impaired cough, or aspiration overwhelm defense.

Observable consequences. Sputum, mucociliary clearance tests, nasal potential difference, microbiology, imaging, cough effectiveness.

Cross-system coupling. Immunity, microbiome, environment, swallowing, nutrition, and occupational exposure determine load and response.

Evidence anchors: [5,10,16]

3.0.8 Immune surveillance and inflammation

Core principle. Resident and recruited immune cells distinguish repairable injury, infection, allergens, and harmful particles.

Control architecture. Barrier signaling, pattern recognition, antibodies, eosinophilic and neutrophilic pathways, resolution mediators, and lymphatics.

Failure modes. Excessive, persistent, misdirected, or suppressed immunity causes asthma, fibrosis, infection, hypersensitivity, or tissue destruction.

Observable consequences. Cell counts, immunoglobulins, FeNO, sputum phenotyping, cultures/PCR, imaging, pathology.

Cross-system coupling. Systemic immunity, medications, age, microbiome, and exposure timing alter phenotype.

Evidence anchors: [5,7,16]

3.0.9 Control of breathing

Core principle. Brainstem rhythm generation, chemoreception, cortical control, sleep networks, and sensory feedback regulate pattern.

Control architecture. Carbon dioxide/pH, oxygen, arousal, voluntary control, exercise command, and mechanoreflexes provide inputs.

Failure modes. Central apnea, unstable loop gain, opioid effect, neurologic injury, congenital control disorders, or maladaptive hyperventilation.

Observable consequences. Polysomnography, capnography, ventilatory response, respiratory inductance, EEG, blood gases.

Cross-system coupling. Neurocognitive, psychiatric, pharmacologic, sleep, cardiac, and altitude states interact.

Evidence anchors: [3,9]

3.0.10 Cough, swallowing, and airway protection

Core principle. Coordinated sensory detection, glottic closure, expiratory force, and swallowing prevent aspiration and clear secretions.

Control architecture. Cranial nerves, brainstem patterning, laryngeal function, cognition, posture, and respiratory muscle force.

Failure modes. Dysphagia, reflux, sedation, neurologic disease, frailty, tracheostomy, or weakness increases aspiration and pneumonia risk.

Observable consequences. Swallow evaluation, cough flow, laryngoscopy, impedance-pH, imaging, aspiration biomarkers.

Cross-system coupling. Neurology, oral health, nutrition, aging, and rehabilitation are integral.

Evidence anchors: [9,21,33]

3.0.11 Exercise and respiratory reserve

Core principle. Exercise increases ventilation, cardiac output, diffusion demand, and respiratory muscle work while requiring efficient gas transport.

Control architecture. Cardiopulmonary coupling, chemoreflexes, peripheral muscle metabolism, conditioning, and pacing determine reserve.

Failure modes. Ventilatory limitation, gas-exchange failure, circulatory limitation, deconditioning, dysfunctional breathing, or anemia limits performance.

Observable consequences. Cardiopulmonary exercise testing, six-minute walk, oxygen uptake, ventilatory equivalents, symptoms, recovery.

Cross-system coupling. Cardiovascular, hematologic, metabolic, musculoskeletal, and psychosocial factors must be separated.

Evidence anchors: [2,8,21]

3.0.12 Development, aging, and repair

Core principle. Airway and alveolar development, peak lung function, injury repair, senescence, and cumulative exposure set lifelong reserve.

Control architecture. Prenatal environment, childhood infection, growth, physical activity, repair programs, and exposure reduction shape trajectory.

Failure modes. Prematurity, impaired growth, recurrent injury, smoking, pollution, fibrosis, and senescence reduce peak or accelerate decline.

Observable consequences. Growth-adjusted spirometry, longitudinal slopes, imaging, diffusion, exercise reserve, functional outcomes.

Cross-system coupling. Maternal-fetal health, life course, environment, immunity, and social determinants are causal bridges.

Evidence anchors: [10,39,40,41]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Alveolar ventilation	$\dot{V}_A = (V_T - V_D)f$	Effective ventilation equals tidal volume minus dead-space volume, multiplied by breathing frequency.	Dead space is heterogeneous and dynamic; minute ventilation alone cannot identify effective gas exchange.
Alveolar carbon dioxide	$P_{ACO_2} \approx K \dot{V}_{CO_2} / \dot{V}_A$	Alveolar carbon dioxide rises with metabolic production and falls with alveolar ventilation.	Assumes steady state and approximate equality between alveolar and arterial carbon dioxide.
Alveolar gas equation	$P_{AO_2} = F_{IO_2}(P_B - P_{H_2O}) - P_{aCO_2}/R$	Estimates alveolar oxygen from inspired oxygen, barometric pressure, water vapor, carbon dioxide, and respiratory quotient.	Requires correct units, inspired fraction, respiratory quotient, and steady-state assumptions.
Fick diffusion	$\dot{V}_{gas} = D A \Delta P / T$	Transfer increases with diffusivity, area, and gradient and decreases with membrane thickness.	The lung is not a uniform membrane; capillary recruitment, reaction with hemoglobin, and transit time matter.
Airway resistance	$R_{aw} = \Delta P / \dot{V}$	Resistance is the pressure difference needed for a given flow.	Flow may be turbulent and airways are collapsible; resistance depends strongly on lung volume and geometry.
Time constant	$\tau = R C$	Regional filling and emptying speed reflects resistance times compliance.	Heterogeneous time constants produce gas trapping and asynchronous ventilation not captured by one global value.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within respiratory health and measurement engineering.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

Continued on next page

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within respiratory health and measurement engineering.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

Evidence anchors: ^[42]

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

Evidence anchors: ^[39,40]

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Evidence anchors: ^[19]

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

Evidence anchors: ^[9]

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

Evidence anchors: ^[22]

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[28]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the respiratory syndrome, disease, failure mode, or functional phenotype, but cross-system processes and functional effects are retained.

5.0.1 Asthma

Definition and phenotype. Variable respiratory symptoms and expiratory airflow limitation arising from heterogeneous airway inflammation, hyperresponsiveness, and remodeling.

Pathophysiology. Epithelial injury, type 2 or non-type 2 inflammation, smooth-muscle constriction, mucus, neural responses, and structural change interact.

Determinants and modifiers. Genetic susceptibility, allergens, viral infection, pollution, tobacco, occupational sensitizers, obesity, stress, medication access, and adherence alter risk and control.

Assessment. History of variability and triggers; objective variable airflow limitation when possible; spirometry/bronchodilator response, peak-flow variability, FeNO or eosinophils in selected contexts; assessment of comorbidity and technique.

Intervention logic. Exposure control and education; inhaled corticosteroid-containing treatment; bronchodilators, phenotype-directed biologics, and action plans according to severity and response.

Known limits and open questions. No single biomarker captures all endotypes; apparent severe asthma may reflect technique, adherence, exposure, vocal dysfunction, or alternative diagnosis.

Suite links. 08 infection/immunity; 09 environment; 14 therapeutics; 05 behavior and anxiety; 03M pregnancy.

Evidence anchors: ^[7,8,10]

5.0.2 Chronic obstructive pulmonary disease

Definition and phenotype. Persistent respiratory symptoms and airflow obstruction associated with airway and/or alveolar abnormalities, usually after cumulative noxious exposure.

Pathophysiology. Small-airway inflammation/remodeling, mucus, emphysematous tissue destruction, loss of recoil, vascular changes, and systemic effects reduce flow and reserve.

Determinants and modifiers. Tobacco, biomass smoke, occupational dust/fumes, pollution, impaired lung growth, infection, socioeconomic conditions, genetics such as alpha-1 antitrypsin deficiency, and aging.

Assessment. Post-bronchodilator spirometry in appropriate symptomatic/exposed people; symptom/exacerbation history, imaging and diffusion when indicated, oxygenation, exercise and comorbidities.

Intervention logic. Exposure cessation, vaccination, activity and pulmonary rehabilitation, inhaled bronchodilators, selected inhaled corticosteroids or other drugs, oxygen for qualifying chronic hypoxemia, exacerbation prevention and palliative support.

Known limits and open questions. Obstruction severity incompletely predicts symptoms, exacerbations, emphysema, mucus disease, pulmonary vascular disease, or functional decline.

Suite links. 03A cardiovascular; 09 environment/occupation; 10 rehabilitation; 07 aging; 14 pharmacology.

Evidence anchors: [6,8,10]

5.0.3 Acute respiratory infection and pneumonia

Definition and phenotype. Infection of airways or lung parenchyma ranging from self-limited disease to hypoxemic failure and sepsis.

Pathophysiology. Pathogen replication, epithelial injury, immune response, alveolar filling, ventilation-perfusion mismatch, systemic inflammation, and secondary organ dysfunction.

Determinants and modifiers. Age, vaccination, immune state, aspiration, crowding, air quality, comorbidity, antimicrobial exposure, pathogen ecology, and health-system access.

Assessment. Syndromic and severity assessment; oxygenation; imaging when it changes management; targeted microbiology based on setting and severity; evaluation for sepsis and complications.

Intervention logic. Supportive care, oxygen/ventilation when needed, timely pathogen-appropriate therapy, source control, infection prevention, vaccination, and stewardship.

Known limits and open questions. Pathogen detection does not always prove causation; radiographic persistence and post-acute symptoms may outlast active infection.

Suite links. 08 infection/AMR; 09 One Health; 11 public health; 12 care delivery.

Evidence anchors: [5,12,13]

5.0.4 Tuberculosis and chronic mycobacterial disease

Definition and phenotype. Airborne mycobacterial infection that may be latent, active, disseminated, drug-resistant, or caused by nontuberculous species.

Pathophysiology. Macrophage infection, granulomatous immunity, tissue necrosis, cavitation, latency/reactivation, and resistance under selective pressure.

Determinants and modifiers. Exposure intensity, crowding, nutrition, HIV/immunosuppression, diabetes, occupational risk, social exclusion, migration, and treatment continuity.

Assessment. Microbiologic confirmation and resistance testing when possible; imaging; immune tests for infection rather than active-disease proof; public-health contact assessment.

Intervention logic. Multidrug therapy, adherence support, toxicity monitoring, infection control, contact management, preventive treatment, and social support.

Known limits and open questions. Latency is a spectrum; immune tests have imperfect predictive value; drug-resistant regimens and implementation remain challenging.

Suite links. 08 infection/immunity; 11 public health; 12 systems; 15 rights/equity.

Evidence anchors: [12,22,42]

5.0.5 Interstitial lung disease and pulmonary fibrosis

Definition and phenotype. A heterogeneous group of inflammatory, fibrotic, occupational, autoimmune, drug-related, genetic, and idiopathic disorders affecting interstitium and alveolar units.

Pathophysiology. Repeated epithelial injury, dysregulated repair, fibroblast activation, extracellular matrix accumulation, architectural distortion, and vascular involvement.

Determinants and modifiers. Age, genetic variants, autoimmune disease, inhalational exposure, drugs, radiation, reflux/aspiration, and unknown triggers.

Assessment. Exposure and autoimmune history, high-resolution CT pattern, spirometry and diffusion, oxygenation/exercise, multidisciplinary review, and selected bronchoscopy/biopsy.

Intervention logic. Remove causes; disease-specific immunomodulation or antifibrotic therapy; oxygen/rehabilitation; transplant evaluation; symptom and palliative support.

Known limits and open questions. Radiologic and pathologic categories can overlap; progression and treatment response vary; invasive diagnosis may not change management.

Suite links. 03H immune; 09 occupation/environment; 14 therapeutics; 10 rehabilitation.

Evidence anchors: [5,8,32]

5.0.6 Pulmonary embolism and chronic thromboembolic disease

Definition and phenotype. Acute or chronic obstruction of pulmonary arteries by thrombus or other material, potentially causing right-heart strain and gas-exchange impairment.

Pathophysiology. Obstruction and vasoconstriction raise pulmonary vascular resistance; dead space and mismatch increase; severe events reduce left-heart filling and systemic perfusion.

Determinants and modifiers. Venous stasis, vascular injury, hypercoagulability, surgery, immobility, cancer, pregnancy, hormones, inflammation, and inherited thrombophilia.

Assessment. Clinical probability, D-dimer in appropriate low/intermediate-risk settings, CT pulmonary angiography or ventilation-perfusion imaging, ultrasound, biomarkers and hemodynamics for severity.

Intervention logic. Anticoagulation, selected reperfusion or thrombectomy, treatment of provoking factors, and evaluation for chronic thromboembolic pulmonary hypertension after persistent symptoms.

Known limits and open questions. Incidental and subsegmental findings require context; prediction tools are population-dependent; post-embolism limitation has multiple causes.

Suite links. 03A cardiovascular; 03G hematology; 03N pregnancy; 04 thrombosis/cancer.

Evidence anchors: [2,30,31]

5.0.7 Pulmonary hypertension

Definition and phenotype. Hemodynamic elevation of pulmonary arterial pressure caused by pulmonary arterial disease, left-heart disease, lung/hypoxia, chronic obstruction, or multifactorial mechanisms.

Pathophysiology. Vascular constriction, remodeling, thrombosis, capillary or venous pathology, elevated left-sided pressure, or loss of vascular bed increases right-ventricular afterload.

Determinants and modifiers. Genetic variants, congenital heart disease, autoimmune disease, drugs/toxins, chronic lung disease, sleep disorders, thromboembolism, liver disease, and infection.

Assessment. Echocardiographic probability and cause evaluation; right-heart catheterization for definitive classification when indicated; exercise, imaging, biomarkers and function.

Intervention logic. Treat underlying cause; specialist phenotype-specific vasodilator strategies; diuresis, oxygen where indicated, exercise/rehabilitation, anticoagulation or surgery for chronic thromboembolic disease.

Known limits and open questions. Noninvasive pressure estimates are imprecise; mixed mechanisms are common; drugs beneficial in one group can harm another.

Suite links. 03A cardiovascular; 03H immunity; 03N maternal; 14 therapeutics.

Evidence anchors: [2,6,9]

5.0.8 Acute respiratory distress syndrome

Definition and phenotype. Acute inflammatory lung injury with bilateral opacities and hypoxemia not fully explained by cardiac failure or fluid overload.

Pathophysiology. Diffuse alveolar injury, permeability edema, surfactant dysfunction, collapse, shunt, microvascular injury, and ventilator-patient interactions.

Determinants and modifiers. Sepsis, pneumonia, aspiration, trauma, pancreatitis, transfusion, inhalation injury, and systemic inflammation.

Assessment. Clinical timing and cause, oxygenation under specified support, imaging, hemodynamic assessment, and exclusion of dominant alternatives.

Intervention logic. Treat cause; lung-protective ventilation, appropriate positive end-expiratory pressure, prone positioning for selected severe disease, conservative fluid strategy after shock, and rescue support when indicated.

Known limits and open questions. Syndrome criteria combine biologically diverse processes; optimal personalization of ventilation and extracorporeal support remains unresolved.

Suite links. 04 critical illness; 08 infection; 12 intensive care; 14 devices/therapeutics.

Evidence anchors: [5,28,29]

5.0.9 Sleep-disordered breathing

Definition and phenotype. Obstructive, central, or mixed events disrupt ventilation, oxygenation, sleep architecture, and autonomic stability.

Pathophysiology. Upper-airway collapse, ventilatory control instability, arousal threshold, anatomic load, neuromuscular control, fluid shift, drugs, or cardiac/neurologic disease.

Determinants and modifiers. Obesity, craniofacial anatomy, age, sex, alcohol/sedatives, nasal obstruction, heart failure, altitude, pregnancy, and neurologic disease.

Assessment. Sleep history and daytime consequences; polysomnography or validated home testing for the intended population; event type, oxygen burden, sleep time and comorbidity.

Intervention logic. Weight and exposure management; positive airway pressure; oral appliances, positional or surgical approaches in selected people; treatment of central causes.

Known limits and open questions. Apnea-hypopnea index is an incomplete phenotype; device algorithms and home tests may misclassify sleep time or event mechanism.

Suite links. 03A cardiovascular; 05 cognition/mood; 07 aging; 03D metabolism.

Evidence anchors: [2,9,19]

5.0.10 Cystic fibrosis and inherited mucociliary disorders

Definition and phenotype. Genetic disorders impair ion transport or ciliary function, producing dehydrated secretions, chronic infection, bronchiectasis, and multi-organ disease.

Pathophysiology. Defective CFTR or ciliary machinery alters airway surface liquid and clearance, driving infection-inflammation cycles and structural damage.

Determinants and modifiers. Genotype, modifier genes, pathogen exposure, nutrition, treatment access, adherence, and environment shape trajectory.

Assessment. Newborn/genetic testing, sweat chloride or functional assays, microbiology, spirometry, imaging, nutrition, and extrapulmonary assessment.

Intervention logic. Airway clearance, antimicrobial strategies, nutrition, pancreatic replacement, highly effective modulator therapy for eligible genotypes, transplant and multidisciplinary care.

Known limits and open questions. Genotype-treatment eligibility and response vary; resistance, cost, fertility, long-term effects, and global access remain major gaps.

Suite links. 02 biology/genomics; 08 microbiome/AMR; 03D/03E metabolism/digestion; 14 pharmacology.

Evidence anchors: ^[12,42]

5.0.11 Bronchiectasis

Definition and phenotype. Permanent bronchial dilation associated with recurrent infection, impaired clearance, inflammation, and diverse underlying causes.

Pathophysiology. A self-reinforcing cycle of impaired clearance, microbial persistence, neutrophilic injury, mucus, and structural distortion.

Determinants and modifiers. Prior infection, immune deficiency, ciliary or CFTR disorders, aspiration, autoimmune disease, obstruction, and idiopathic factors.

Assessment. High-resolution CT, sputum microbiology, exacerbation history, lung function, cause-directed laboratory evaluation, and functional assessment.

Intervention logic. Airway clearance, vaccination, cause treatment, targeted antibiotics, selected anti-inflammatory strategies, rehabilitation, and management of hemoptysis.

Known limits and open questions. Microbiome signatures and inflammatory phenotypes do not yet define universally effective precision treatment.

Suite links. 08 infection/immunity; 03H immune deficiency; 03L aspiration/oral; 14 antimicrobials.

Evidence anchors: ^[6,12,16]

5.0.12 Pleural disease

Definition and phenotype. Air, fluid, blood, pus, malignant tissue, or fibrosis in the pleural space can impair mechanics and signal systemic disease.

Pathophysiology. Pleural pressure uncoupling, inflammation, capillary/lymphatic imbalance, infection, malignancy, trauma, or fistula changes lung expansion.

Determinants and modifiers. Infection, heart or kidney failure, cancer, embolism, surgery, trauma, asbestos, autoimmune disease, and ventilation.

Assessment. Ultrasound and radiography/CT, fluid analysis when indicated, manometry in selected contexts, cytology/microbiology and pleural biopsy for unresolved causes.

Intervention logic. Observe or drain according to physiology and cause; antibiotics, pleurodesis, surgery, malignancy treatment, and symptom palliation.

Known limits and open questions. Fluid biomarkers and cytology have imperfect sensitivity; intervention burden must be weighed against symptoms and prognosis.

Suite links. 03A cardiovascular; 03F renal; 04 cancer; 09 occupational exposure.

Evidence anchors: ^[5]

5.0.13 Pneumothorax and air leak

Definition and phenotype. Air in the pleural space causes partial or complete lung collapse; tension physiology can obstruct venous return and threaten life.

Pathophysiology. Rupture of subpleural structures, trauma, procedure, pressure injury, or fistula permits gas entry faster than removal.

Determinants and modifiers. Underlying lung disease, smoking, body habitus, mechanical ventilation, trauma, procedures, and genetic syndromes.

Assessment. Clinical stability and ultrasound/radiography; CT for selected recurrent or complex disease; evaluate tension physiology immediately.

Intervention logic. Emergency decompression when unstable; observation, aspiration, tube drainage, surgery or pleurodesis according to size, cause, leak, recurrence and resources.

Known limits and open questions. Size measures differ by method; recurrence prediction is imprecise; ambulatory devices require careful selection and follow-up.

Suite links. 04 trauma; 12 acute care; 14 procedures/devices.

Evidence anchors: ^[5,28]

5.0.14 Occupational and environmental lung disease

Definition and phenotype. Acute or chronic respiratory injury caused or amplified by workplace and ambient exposures.

Pathophysiology. Particles, fibers, fumes, gases, allergens, heat, combustion products, and sensitizers trigger inflammation, fibrosis, cancer, bronchoconstriction, or systemic effects.

Determinants and modifiers. Dose, particle chemistry/size, ventilation, controls, protective equipment, coexposures, socioeconomic position, climate, and susceptibility.

Assessment. Complete job and exposure history, temporal symptom pattern, spirometry/peak flow, imaging, biomarkers or challenge testing in selected settings, industrial hygiene data.

Intervention logic. Elimination/substitution and engineering controls first; exposure cessation, compensation/accommodation, disease-specific care, surveillance, and public policy.

Known limits and open questions. Personal sensors may not reflect biologically effective dose; latency and mixed exposures complicate attribution; worker protections vary.

Suite links. 09 environment/occupation; 11 prevention; 15 law/equity; 04 cancer/fibrosis.

Evidence anchors: ^[10,11]

5.0.15 Lung cancer

Definition and phenotype. Malignant epithelial tumors of lung and airways with diverse molecular drivers, histology, stage, treatment response, and systemic effects.

Pathophysiology. Cumulative genomic/epigenomic injury, clonal selection, immune evasion, invasion, angiogenesis, and metastasis.

Determinants and modifiers. Tobacco, radon, occupational carcinogens, pollution, prior radiation, family/genetic factors, and stochastic processes.

Assessment. Risk-based screening where recommended; diagnostic imaging, tissue/cytology, molecular profiling, staging, performance status, pulmonary reserve, and goals.

Intervention logic. Prevention and cessation; surgery, radiation, systemic targeted, immune or cytotoxic therapy by stage/biology; symptom and palliative care.

Known limits and open questions. Screening benefit depends on eligibility and implementation; incidental nodules and overdiagnosis create harm; resistance is expected.

Suite links. 04 cancer; 09 environment; 14 therapeutics; 10 function/palliative care.

Evidence anchors: ^[10,17,18]

5.0.16 Post-acute respiratory syndromes and long COVID

Definition and phenotype. Persistent breathlessness, cough, exercise intolerance, dysautonomia, fatigue, or imaging/function abnormalities after acute infection.

Pathophysiology. Potential mechanisms include residual injury, vascular/endothelial change, immune dysregulation, autonomic dysfunction, dysfunctional breathing, deconditioning, and comorbidity unmasking.

Determinants and modifiers. Acute severity, prior health, repeated infection, vaccination, social determinants, mental health, and access may modify risk.

Assessment. Symptom- and safety-directed assessment: oxygenation, spirometry/diffusion, imaging or cardiopulmonary testing when indicated; screen cardiac, thrombotic, autonomic, neurologic and functional domains.

Intervention logic. Treat identified disease; pacing or graded rehabilitation individualized to post-exertional response; symptom management, vaccination and reinfection prevention, coordinated care.

Known limits and open questions. No single biomarker or mechanism explains all cases; treatment evidence and subgroup definitions remain incomplete.

Suite links. 04 systemic/post-acute; 05 cognition; 10 rehabilitation; 08 immunity; 03A cardiovascular.

Evidence anchors: ^[15,20,24]

5.0.17 Respiratory failure

Definition and phenotype. Failure to maintain adequate oxygenation, carbon-dioxide clearance, or both, acutely or chronically.

Pathophysiology. Gas-exchange failure, pump failure, airway obstruction, excessive demand, dead space, reduced inspired oxygen, or combinations overwhelm reserve.

Determinants and modifiers. Underlying disease, sedatives/opioids, obesity, neuromuscular weakness, infection, fluid overload, altitude, and treatment delay.

Assessment. Urgent clinical assessment; pulse oximetry plus blood gas when ventilation/acid-base matters; cause localization; work of breathing, mental status, imaging and hemodynamics.

Intervention logic. Oxygen titrated to context, noninvasive or invasive ventilatory support, cause treatment, secretion management, and goals-of-care planning.

Known limits and open questions. Oxygen saturation alone cannot detect hypercapnia or tissue oxygen delivery; thresholds depend on chronic adaptation and cause.

Suite links. 12 acute care; 14 devices; 03C neuromuscular; 03A hemodynamics.

Evidence anchors: ^[3,8,28]

Building a multimodal respiratory monitoring platform

A useful platform begins with clinical questions, not sensor count. It should preserve synchronized raw signals, context, quality and device provenance so that respiratory patterns can be distinguished from motion, cardiac pulsation, speech, sleep state and environment.

6.1 Tier 1: low-risk longitudinal platform

- Validated pulse oximeter with accessible waveform and signal quality.
- Respiratory inductance or stretch sensing with explicit posture and motion channels.
- Capnography when the intended claim requires ventilation rather than only oxygenation.
- Portable calibrated spirometry for scheduled testing rather than continuous inference.
- Environmental temperature, humidity, carbon dioxide and particulate measurements with placement metadata.
- Symptom, medication, exposure, sleep, activity and infection event logging.
- Common clock, raw export, quality flags and immutable device/firmware identifiers.

Selected evidence anchors: [8,11,23]

6.2 Tier 2: challenge and reference laboratory

- Metabolic cart and cardiopulmonary exercise system.
- Full pulmonary function laboratory: spirometry, lung volumes, diffusion and respiratory pressures.
- Reference airflow, gas analyzers and blood gases as indicated.
- Polysomnography or high-fidelity sleep respiratory channels.
- Ultrasound and clinical imaging interfaces; calibration phantoms.
- Trained clinical supervision, emergency response and prespecified stop criteria.

Selected evidence anchors: [8,9,27]

6.3 Claims that should remain separate

Signal	Defensible narrow claim	Unsafe overreach
SpO2	Estimate arterial saturation within validated conditions.	Infer ventilation, tissue oxygen delivery, cause of hypoxemia, or absence of respiratory disease.
Chest motion	Estimate breathing timing or relative pattern under validated posture/activity.	Infer tidal volume or disease without calibration and artifact channels.

Continued on next page

Signal	Defensible narrow claim	Unsafe overreach
Microphone	Detect selected cough/acoustic events in validated environments.	Diagnose infection, asthma or COPD from sound alone.
Portable spirometry	Measure forced flow/volume when quality criteria are met.	Interpret poor-quality blows as decline or rule out small-airway/parenchymal disease.
Environmental sensor	Characterize local proxy exposure at sensor location.	Assign individual biologic dose or causation without time-activity and calibration.

Selected evidence anchors: [8,11,24]

Reference methods, calibration, and artifact challenges

Respiratory devices fail in predictable ways. Validation must cover the intended physiological range, interfaces, skin and body characteristics, activity, environment, disease severity and operator skill rather than only quiet healthy volunteers.

7.1 Minimum bench tests

- Traceable flow and volume waveforms across rates and gas composition.
- Known carbon-dioxide and oxygen mixtures, humidity and temperature.
- Leak, occlusion, condensation, secretion and interface-displacement testing.
- Mechanical chest simulators with variable frequency, amplitude, posture and superimposed motion.
- Power interruption, clock drift, packet loss, saturation and recovery.
- Cleaning, contamination, disposables, sensor aging and lot-to-lot variation.

Selected evidence anchors: [25,26,27]

7.2 Minimum human validation strata

- Healthy and disease participants spanning the intended range, including low perfusion or severe obstruction when claimed.
- Skin pigmentation and characteristics relevant to optical sensing.
- Children, pregnancy, older age or disability only when explicitly intended and ethically justified.
- Rest, speech, cough, sleep, ambulation, exercise and real-world home conditions.
- Reference measurements collected close enough in time and under equivalent physiological state.
- Prespecified handling of failed, rejected and missing measurements.

Selected evidence anchors: [22,24,44]

Respiratory emergencies and red flags

A monitoring system must not create false reassurance or delay emergency care. Algorithms should prioritize reliable detection of gross deterioration, signal failure and unsafe use over speculative diagnosis.

8.1 Urgent patterns

- Severe or rapidly worsening breathlessness, inability to speak, exhaustion, altered mental status or cyanosis.
- Marked hypoxemia or a falling trend confirmed with signal-quality review and clinical assessment.
- Silent chest, stridor, suspected anaphylaxis, airway foreign body or tension pneumothorax.
- Hemoptysis with instability, suspected pulmonary embolism, severe chest pain or circulatory compromise.
- Rising carbon dioxide/acidemia, apnea or opioid/sedative exposure.
- Device alarms, disconnection, occlusion, fire risk or delayed escalation.

Selected evidence anchors: [28,29]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current respiratory health and measurement engineering risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

9.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current respiratory health and measurement engineering risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

9.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.
Evidence anchors: ^[21]

9.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.
Evidence anchors: ^[3,5]

9.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

9.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

Evidence anchors: ^[21]

9.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

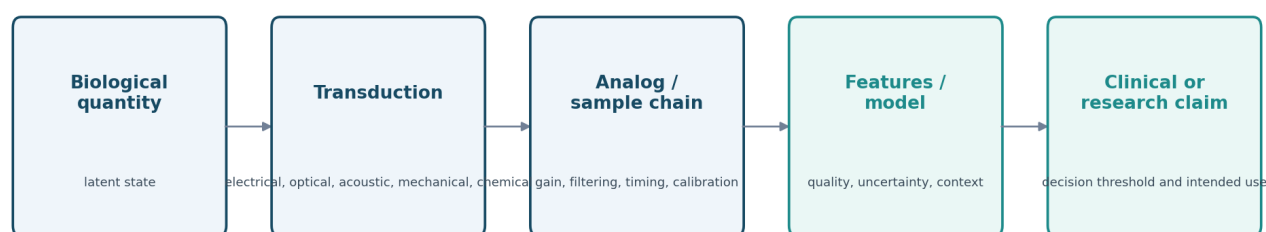
Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Evidence anchors: ^[43]

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 10.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
History, observation, and respiratory rate	Noninvasive	Symptoms, speech, posture, respiratory timing, accessory-muscle use, cough, cyanosis and context.	Urgency, pattern, triggers, functional impact and hypotheses.	Immediate, low cost, longitudinally meaningful.	Rate is often measured inaccurately; symptoms are multidetermined and can be absent in severe disease.
Pulse oximetry	Noninvasive optical	Red/infrared photoplethysmography estimates arterial oxygen saturation and pulse.	SpO2, pulse rate, plethysmographic waveform and signal-quality indicators.	Continuous, inexpensive, portable and useful for hypoxemia detection.	Motion, low perfusion, dyshemoglobins, pigmentation-related bias, nail coatings, venous pulsation and device algorithms; does not measure ventilation.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Capnography	Noninvasive to minimally invasive depending interface	Infrared carbon-dioxide absorption in exhaled gas.	End-tidal CO ₂ , waveform, respiratory rate, dead-space and ventilation clues.	Rapid detection of apnea, disconnection and ventilation trend; waveform localizes some problems.	Sampling delay, leaks, dilution, mouth breathing, oxygen flow and severe mismatch make end-tidal differ from arterial CO ₂ .
Spirometry and flow-volume loops	Noninvasive effort-dependent	Pneumotachograph, turbine, ultrasonic or displacement sensor measures volume and flow.	FEV ₁ , FVC, ratio, loop morphology, bronchodilator response and trends.	Core test for airflow obstruction; portable and standardized.	Technique, effort, leaks, cough, reference equations and device calibration; normal spirometry does not exclude disease.
Peak expiratory flow	Noninvasive effort-dependent	Mechanical or electronic flow meter captures maximal expiratory flow.	Daily variability, personal best, action-plan thresholds.	Low cost and useful for longitudinal asthma self-management.	Strongly effort- and device-dependent; poor technique and different meters limit interchangeability.
Lung volumes by plethysmography or gas dilution	Noninvasive laboratory	Pressure/volume relation in body box or dilution/washout of inert gas.	TLC, RV, FRC, gas trapping and airway resistance depending method.	Differentiates restriction, hyperinflation and trapped gas.	Methods measure different compartments; leaks, panting technique, severe obstruction and equipment complexity.
Diffusing capacity	Noninvasive breath maneuver	Transfer of carbon monoxide, sometimes nitric oxide, estimates gas-transfer conductance.	DLCO and components adjusted for relevant conditions.	Sensitive to emphysema, interstitial and pulmonary vascular disease and anemia correction.	Depends on hemoglobin, inspired volume, carboxyhemoglobin, altitude, technique and capillary blood volume.
Impulse oscillometry / forced oscillation	Noninvasive passive breathing	External pressure oscillations measure respiratory impedance over frequency.	Resistance, reactance, frequency dependence and small-airway clues.	Requires less forced effort and can assess tidal breathing.	Upper-airway shunt, cheek support, device/protocol differences and uncertain treatment thresholds.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Multiple-breath washout	Noninvasive	Inert gas washout tracks ventilation heterogeneity.	Lung clearance index, functional residual capacity and regional inhomogeneity proxies.	Sensitive to early small-airway disease in selected populations.	Long tests, leaks, gas/analyzer differences, dead-space corrections and limited universal reference ranges.
Exhaled nitric oxide	Noninvasive breath test	Electrochemical or chemiluminescent sensing of flow-standardized exhaled NO.	FeNO as a proxy for selected type 2 airway inflammation and steroid response likelihood.	Rapid and useful as one component of asthma phenotyping.	Affected by flow, age, atopy, infection, smoking, diet and treatment; low values do not exclude asthma.
Arterial, venous, and transcutaneous blood gases	Minimally invasive to invasive	Electrochemical/optical measurement of pH, oxygen and carbon dioxide in blood or through heated skin.	Acid-base state, PaO ₂ , PaCO ₂ , bicarbonate and co-oximetry where available.	Reference for ventilation and oxygenation in acute care; reveals hypercapnia missed by oximetry.	Sampling error; air contamination, delay, temperature, venous-arterial differences and transcutaneous drift.
Chest radiography	Noninvasive ionizing imaging	X-ray attenuation projection.	Large-scale structure, opacity, volume, pleural air/fluid, device position.	Fast, widely available and useful in acute care.	Two-dimensional overlap, limited sensitivity/specificity and radiation; normal image does not exclude important disease.
Computed tomography and CT angiography	Noninvasive to minimally invasive with contrast; ionizing	Tomographic X-ray attenuation with high spatial resolution.	Parenchyma, airways, nodules, emphysema, fibrosis, vasculature and quantitative features.	High structural detail and rapid angiography.	Radiation, contrast risk, incidental findings, protocol dependence and algorithmic measurement drift.
Lung ultrasound	Noninvasive acoustic	Reflected ultrasound from pleura, tissue interfaces and fluid.	Pleural sliding, pneumothorax signs, interstitial artifact patterns, consolidation and effusion.	Portable, repeatable and radiation-free at bedside.	Air limits direct parenchymal imaging; artifacts are nonspecific and operator/protocol dependent.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Bronchoscopy and endobronchial ultrasound	Invasive	Optical airway visualization, ultrasound, lavage, brushings, biopsy and intervention.	Anatomy, obstruction, infection/cytology specimens, lymph-node or lesion sampling.	Direct access and combined diagnostic/therapeutic capability.	Sedation, bleeding, pneumothorax, sampling error, contamination and selection bias.
Polysomnography and home sleep apnea testing	Noninvasive multichannel	EEG/EOG/EMG, airflow, effort, oximetry, ECG, position and audio; home systems use subsets.	Sleep stages, event type/index, oxygen burden, arousals and timing.	Reference characterization of sleep and breathing interaction.	Home tests may misestimate sleep time and miss nonobstructive disorders; scoring and sensor loss vary.
Cardiopulmonary exercise testing	Noninvasive to minimally invasive	Breath-by-breath gas analysis integrated with ECG, pressure, oximetry and workload; invasive versions add pressures/gases.	VO ₂ , VCO ₂ , ventilatory efficiency, thresholds, reserve and limitation pattern.	Integrates respiratory, cardiac, vascular and metabolic responses under challenge.	Requires expertise; effort, protocol and interpretation are multidimensional; rare adverse events.
Respiratory inductance, impedance pneumography and wearable motion sensing	Noninvasive	Chest/abdominal displacement, thoracic impedance or inertial sensing estimates respiratory timing and effort.	Rate, pattern, relative effort, apnea candidates, activity context.	Continuous and feasible outside laboratory.	Posture, motion, belt placement, anatomy, speech and calibration distort tidal volume and event classification.
Sputum, breath condensate, volatile and aerosol analysis	Noninvasive sample collection	Cells, proteins, nucleic acids, metabolites, microbes, particles and volatile compounds.	Inflammatory, infectious or exposure signatures and research phenotypes.	Direct airway-derived material and repeatability potential.	Oral contamination, variable dilution, unstable compounds, sampling induction effects and weak clinical thresholds.

10.0.1 History, observation, and respiratory rate

Measured quantity and transducer. Symptoms, speech, posture, respiratory timing, accessory-muscle use, cough, cyanosis and context.

Typical outputs. Urgency, pattern, triggers, functional impact and hypotheses.

Strengths. Immediate, low cost, longitudinally meaningful.

Limitations and artifacts. Rate is often measured inaccurately; symptoms are multidetermined and can be absent in severe disease.

Reference and validation strategy. Standardized counting interval, repeated measures, observer training, linkage to outcomes.

Clinical/research maturity. Mature clinical method.

Evidence anchors: [3,21]

10.0.2 Pulse oximetry

Measured quantity and transducer. Red/infrared photoplethysmography estimates arterial oxygen saturation and pulse.

Typical outputs. SpO₂, pulse rate, plethysmographic waveform and signal-quality indicators.

Strengths. Continuous, inexpensive, portable and useful for hypoxemia detection.

Limitations and artifacts. Motion, low perfusion, dyshemoglobins, pigmentation-related bias, nail coatings, venous pulsation and device algorithms; does not measure ventilation.

Reference and validation strategy. Bench optical standards plus arterial co-oximetry across range, motion, perfusion, skin characteristics and intended setting.

Clinical/research maturity. Mature, with important performance limits.

Evidence anchors: [22,23]

10.0.3 Capnography

Measured quantity and transducer. Infrared carbon-dioxide absorption in exhaled gas.

Typical outputs. End-tidal CO₂, waveform, respiratory rate, dead-space and ventilation clues.

Strengths. Rapid detection of apnea, disconnection and ventilation trend; waveform localizes some problems.

Limitations and artifacts. Sampling delay, leaks, dilution, mouth breathing, oxygen flow and severe mismatch make end-tidal differ from arterial CO₂.

Reference and validation strategy. Gas standards, response time, humidity, flow, interface and arterial comparison in target population.

Clinical/research maturity. Mature in anesthesia/acute care; expanding ambulatory use.

Evidence anchors: [3,27]

10.0.4 Spirometry and flow-volume loops

Measured quantity and transducer. Pneumotachograph, turbine, ultrasonic or displacement sensor measures volume and flow.

Typical outputs. FEV1, FVC, ratio, loop morphology, bronchodilator response and trends.

Strengths. Core test for airflow obstruction; portable and standardized.

Limitations and artifacts. Technique, effort, leaks, cough, reference equations and device calibration; normal spirometry does not exclude disease.

Reference and validation strategy. ATS/ERS acceptability/repeatability, calibration verification, biological controls and appropriate reference models.

Clinical/research maturity. Mature reference method for defined claims.

Evidence anchors: [6,7,8]

10.0.5 Peak expiratory flow

Measured quantity and transducer. Mechanical or electronic flow meter captures maximal expiratory flow.

Typical outputs. Daily variability, personal best, action-plan thresholds.

Strengths. Low cost and useful for longitudinal asthma self-management.

Limitations and artifacts. Strongly effort- and device-dependent; poor technique and different meters limit interchangeability.

Reference and validation strategy. Flow simulator, within-device repeatability, technique training and individualized baseline.

Clinical/research maturity. Mature narrow-use tool.

Evidence anchors: [7,8]

10.0.6 Lung volumes by plethysmography or gas dilution

Measured quantity and transducer. Pressure/volume relation in body box or dilution/washout of inert gas.

Typical outputs. TLC, RV, FRC, gas trapping and airway resistance depending method.

Strengths. Differentiates restriction, hyperinflation and trapped gas.

Limitations and artifacts. Methods measure different compartments; leaks, panting technique, severe obstruction and equipment complexity.

Reference and validation strategy. Physical calibration, protocol standardization, repeatability and cross-method interpretation.

Clinical/research maturity. Mature laboratory testing.

Evidence anchors: [3,8]

10.0.7 Diffusing capacity

Measured quantity and transducer. Transfer of carbon monoxide, sometimes nitric oxide, estimates gas-transfer conductance.

Typical outputs. DLCO and components adjusted for relevant conditions.

Strengths. Sensitive to emphysema, interstitial and pulmonary vascular disease and anemia correction.

Limitations and artifacts. Depends on hemoglobin, inspired volume, carboxyhemoglobin, altitude, technique and capillary blood volume.

Reference and validation strategy. Gas and volume calibration, standard timing, hemoglobin adjustment and reference equations.

Clinical/research maturity. Mature but model-dependent.

Evidence anchors: [3,8]

10.0.8 Impulse oscillometry / forced oscillation

Measured quantity and transducer. External pressure oscillations measure respiratory impedance over frequency.

Typical outputs. Resistance, reactance, frequency dependence and small-airway clues.

Strengths. Requires less forced effort and can assess tidal breathing.

Limitations and artifacts. Upper-airway shunt, cheek support, device/protocol differences and uncertain treatment thresholds.

Reference and validation strategy. Mechanical impedance loads, repeatability, reference data and comparison to clinical outcomes.

Clinical/research maturity. Established adjunct; not universal replacement for spirometry.

Evidence anchors: [7,8]

10.0.9 Multiple-breath washout

Measured quantity and transducer. Inert gas washout tracks ventilation heterogeneity.

Typical outputs. Lung clearance index, functional residual capacity and regional inhomogeneity proxies.

Strengths. Sensitive to early small-airway disease in selected populations.

Limitations and artifacts. Long tests, leaks, gas/analyzer differences, dead-space corrections and limited universal reference ranges.

Reference and validation strategy. Cross-device standardization, known-volume lung models, repeatability and longitudinal outcome studies.

Clinical/research maturity. Specialized / context-limited.

Evidence anchors: ^[8]

10.0.10 Exhaled nitric oxide

Measured quantity and transducer. Electrochemical or chemiluminescent sensing of flow-standardized exhaled NO.

Typical outputs. FeNO as a proxy for selected type 2 airway inflammation and steroid response likelihood.

Strengths. Rapid and useful as one component of asthma phenotyping.

Limitations and artifacts. Affected by flow, age, atopy, infection, smoking, diet and treatment; low values do not exclude asthma.

Reference and validation strategy. Gas calibration, flow control, repeatability and claim-specific outcome validation.

Clinical/research maturity. Mature adjunct with limited scope.

Evidence anchors: ^[7,8]

10.0.11 Arterial, venous, and transcutaneous blood gases

Measured quantity and transducer. Electrochemical/optical measurement of pH, oxygen and carbon dioxide in blood or through heated skin.

Typical outputs. Acid-base state, PaO₂, PaCO₂, bicarbonate and co-oximetry where available.

Strengths. Reference for ventilation and oxygenation in acute care; reveals hypercapnia missed by oximetry.

Limitations and artifacts. Sampling error, air contamination, delay, temperature, venous-arterial differences and transcutaneous drift.

Reference and validation strategy. Analyzer QC, proficiency testing, preanalytic protocol and method-specific comparison.

Clinical/research maturity. Mature clinical laboratory method.

Evidence anchors: [3,27]

10.0.12 Chest radiography

Measured quantity and transducer. X-ray attenuation projection.

Typical outputs. Large-scale structure, opacity, volume, pleural air/fluid, device position.

Strengths. Fast, widely available and useful in acute care.

Limitations and artifacts. Two-dimensional overlap, limited sensitivity/specificity and radiation; normal image does not exclude important disease.

Reference and validation strategy. Equipment QA, dose optimization, blinded interpretation and outcome-linked appropriateness.

Clinical/research maturity. Mature.

Evidence anchors: [5,28]

10.0.13 Computed tomography and CT angiography

Measured quantity and transducer. Tomographic X-ray attenuation with high spatial resolution.

Typical outputs. Parenchyma, airways, nodules, emphysema, fibrosis, vasculature and quantitative features.

Strengths. High structural detail and rapid angiography.

Limitations and artifacts. Radiation, contrast risk, incidental findings, protocol dependence and algorithmic measurement drift.

Reference and validation strategy. Phantom and dose QA, protocol harmonization, segmentation reproducibility and clinical reference correlation.

Clinical/research maturity. Mature; quantitative biomarkers need claim-specific validation.

Evidence anchors: [17,24]

10.0.14 Lung ultrasound

Measured quantity and transducer. Reflected ultrasound from pleura, tissue interfaces and fluid.

Typical outputs. Pleural sliding, pneumothorax signs, interstitial artifact patterns, consolidation and effusion.

Strengths. Portable, repeatable and radiation-free at bedside.

Limitations and artifacts. Air limits direct parenchymal imaging; artifacts are nonspecific and operator/protocol dependent.

Reference and validation strategy. Phantoms, standardized zones, observer agreement, reference imaging and decision-impact studies.

Clinical/research maturity. Mature point-of-care adjunct.

Evidence anchors: ^[27,28]

10.0.15 Bronchoscopy and endobronchial ultrasound

Measured quantity and transducer. Optical airway visualization, ultrasound, lavage, brushings, biopsy and intervention.

Typical outputs. Anatomy, obstruction, infection/cytology specimens, lymph-node or lesion sampling.

Strengths. Direct access and combined diagnostic/therapeutic capability.

Limitations and artifacts. Sedation, bleeding, pneumothorax, sampling error, contamination and selection bias.

Reference and validation strategy. Procedure standards, specimen adequacy, pathology/microbiology reference and complication surveillance.

Clinical/research maturity. Mature specialist procedure.

Evidence anchors: ^[28]

10.0.16 Polysomnography and home sleep apnea testing

Measured quantity and transducer. EEG/EOG/EMG, airflow, effort, oximetry, ECG, position and audio; home systems use subsets.

Typical outputs. Sleep stages, event type/index, oxygen burden, arousals and timing.

Strengths. Reference characterization of sleep and breathing interaction.

Limitations and artifacts. Home tests may misestimate sleep time and miss nonobstructive disorders; scoring and sensor loss vary.

Reference and validation strategy. AASM protocols, sensor checks, scorer reliability, device-specific validation and clinical outcome assessment.

Clinical/research maturity. Mature for defined indications.

Evidence anchors: [9]

10.0.17 Cardiopulmonary exercise testing

Measured quantity and transducer. Breath-by-breath gas analysis integrated with ECG, pressure, oximetry and workload; invasive versions add pressures/gases.

Typical outputs. VO₂, VCO₂, ventilatory efficiency, thresholds, reserve and limitation pattern.

Strengths. Integrates respiratory, cardiac, vascular and metabolic responses under challenge.

Limitations and artifacts. Requires expertise; effort, protocol and interpretation are multidimensional; rare adverse events.

Reference and validation strategy. Metabolic calibration, protocol standardization, repeatability and mechanistic reference testing.

Clinical/research maturity. Mature specialized test.

Evidence anchors: [2,8]

10.0.18 Respiratory inductance, impedance pneumography and wearable motion sensing

Measured quantity and transducer. Chest/abdominal displacement, thoracic impedance or inertial sensing estimates respiratory timing and effort.

Typical outputs. Rate, pattern, relative effort, apnea candidates, activity context.

Strengths. Continuous and feasible outside laboratory.

Limitations and artifacts. Posture, motion, belt placement, anatomy, speech and calibration distort tidal volume and event classification.

Reference and validation strategy. Bench motion rigs, simultaneous reference airflow/effort, free-living artifact and subgroup testing.

Clinical/research maturity. Research to established adjunct depending claim.

Evidence anchors: [23,24]

10.0.19 Sputum, breath condensate, volatile and aerosol analysis

Measured quantity and transducer. Cells, proteins, nucleic acids, metabolites, microbes, particles and volatile compounds.

Typical outputs. Inflammatory, infectious or exposure signatures and research phenotypes.

Strengths. Direct airway-derived material and repeatability potential.

Limitations and artifacts. Oral contamination, variable dilution, unstable compounds, sampling induction effects and weak clinical thresholds.

Reference and validation strategy. Preanalytic standards, blanks, recovery studies, reference assays, locked validation and utility trials.

Clinical/research maturity. Mostly specialized or frontier beyond routine microbiology.

Evidence anchors: ^[16,43]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise respiratory health and measurement engineering quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

11.0.1 Intended use and measurand

Design requirements. Define the precise respiratory health and measurement engineering quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: [23,26]

11.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: [26,27]

11.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

Evidence anchors: ^[43]

11.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[25]

11.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[25,43]

11.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

Evidence anchors: [24]

11.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: [24]

11.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: [27,44]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Tobacco and inhaled-exposure prevention	Upstream and individual prevention	Eliminate combustible tobacco and reduce secondhand smoke, vaping aerosols, biomass, radon, occupational and ambient exposures.	Large causal evidence supports exposure reduction; policy, engineering control and cessation support outperform advice alone.	Withdrawal, unequal access to treatment, substitution with unvalidated products, economic and housing constraints.	Exposure history, biochemical verification when useful, symptoms, lung function, in-door/workplace measures and policy reach.
Vaccination and infection prevention	Primary prevention	Reduce respiratory infection, severe disease and transmission through immunization, ventilation, hygiene and risk-specific precautions.	Strong but pathogen-, product-, age- and season-specific evidence; benefits include indirect protection.	Rare adverse events, waning, mismatch, access, hesitancy and inequitable distribution.	Coverage, breakthrough severity, adverse-event surveillance, ventilation/air metrics and equity.
Inhaled corticosteroid-containing therapy	Disease-modifying pharmacology	Suppress airway inflammation; central to asthma control and selected COPD phenotypes.	Reduces asthma exacerbation risk; benefit in COPD depends on exacerbation and inflammatory phenotype with pneumonia tradeoff.	Local candidiasis/dysphonia, systemic exposure at high dose, pneumonia risk in selected COPD, technique and adherence failures.	Symptoms, exacerbations, reliever use, spirometry, eosinophilic markers where useful, dose and technique.
Bronchodilators	Symptom and airflow pharmacology	Relax airway smooth muscle through beta-adrenergic or muscarinic pathways.	Improve symptoms and function in obstructive disease; regimen depends on diagnosis and severity.	Tremor, tachycardia, urinary/ocular effects, paradoxical bronchospasm, overreliance and device mismatch.	Symptoms, rescue use, heart rate, spirometry, side effects, inhaler technique and adherence.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Biologic and precision respiratory therapy	Targeted pharmacology	Block defined immune pathways or correct specific molecular defects such as eligible CFTR variants.	Large benefit in selected biomarker/genotype-defined groups; not evidence that all patients with the label share the target.	Cost, injection/infusion burden, infection or immune risks, access disparity, nonresponse and uncertain duration.	Eligibility criteria, exacerbations, function, biomarkers, adverse events, adherence and de-escalation evidence.
Antimicrobial treatment and stewardship	Etiologic treatment and system intervention	Treat likely or confirmed bacterial, mycobacterial, fungal or viral disease while minimizing resistance and collateral harm.	Timely appropriate therapy is lifesaving in severe infection; unnecessary exposure causes harm and resistance.	Allergy, toxicity, interactions, microbiome disruption, Clostridioides difficile, resistance and diagnostic uncertainty.	Clinical trajectory, cultures/PCR and susceptibilities, drug levels/organ function when needed, adverse events and local resistance.
Oxygen therapy	Supportive therapy	Increase inspired oxygen to correct clinically important hypoxemia.	Essential in acute hypoxemia and improves survival in selected chronic severe resting hypoxemia; benefit is not universal for breathlessness without hypoxemia.	Hyperoxia, carbon-dioxide retention in susceptible patients, fire risk, dryness, mobility burden and inequitable supply.	Target saturation by condition, blood gases when ventilation matters, flow/device performance, adherence and functional benefit.
Noninvasive ventilation and positive airway pressure	Mechanical support	Apply positive pressure to splint airway, unload muscles, improve ventilation or recruit lung.	Established for obstructive sleep apnea and selected acute/chronic ventilatory failure; benefit depends on cause, interface and adherence.	Aspiration, skin injury, leaks, intolerance, hemodynamic effects, delayed intubation and algorithm error.	Gas exchange, work of breathing, synchrony, leak, adherence, pressure data, skin and patient goals.
Invasive mechanical ventilation	Acute life support	Provide controlled or assisted ventilation and oxygenation through an artificial airway.	Life-saving when failure cannot be supported otherwise; lung-protective approaches reduce ventilator-induced injury in defined syndromes.	Barotrauma/volutrauma, diaphragm weakness, pneumonia, delirium, sedation harm, hemodynamic compromise and long-term disability.	Pressures/volumes, blood gases, synchrony, sedation, infection, diaphragm and mobility, readiness to liberate.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Pulmonary rehabilitation	Rehabilitation	Combine exercise training, education, behavior support, breathing/energy strategies and multidisciplinary management.	Improves exercise capacity, symptoms and quality of life in chronic respiratory disease and after selected exacerbations.	Access, transportation, desaturation risk, comorbidity, inadequate personalization and poor maintenance support.	Functional tests, symptoms, participation, oxygen response, adherence, goals and adverse events.
Airway clearance and secretion management	Self-management and rehabilitation	Mobilize and remove secretions through hydration, oscillation, positive expiratory pressure, positioning, cough assistance or suction.	Established in cystic fibrosis and selected bronchiectasis/neuromuscular phenotypes; technique should match mechanism.	Fatigue, reflux/aspiration, desaturation, hemoptysis, device contamination and treatment burden.	Sputum burden, exacerbations, technique, cough flow, oxygenation and patient-reported feasibility.
Surgery, endoscopy and transplantation	Procedural/definitive treatment	Resect, repair, drain, reduce volume, reopen airway, remove thrombus or replace failing lungs in selected disease.	Can be curative or transformative when anatomy, stage, reserve and goals align.	Perioperative death, bleeding, infection, rejection, lifelong immunosuppression, access and opportunity cost.	Multidisciplinary selection, pulmonary/cardiac reserve, pathology, complications, function, graft status and survival.
Palliative and supportive respiratory care	Whole-person supportive care	Treat breathlessness, cough, anxiety, secretion distress, fatigue and caregiver burden while aligning treatment with goals.	Symptom-focused approaches improve experience and can coexist with disease-directed therapy.	Sedation, respiratory depression, stigma, fragmented access and late referral.	Symptoms, function, goals, caregiver burden, adverse effects and crisis plans.

12.0.1 Tobacco and inhaled-exposure prevention

Target and mechanism. Eliminate combustible tobacco and reduce secondhand smoke, vaping aerosols, biomass, radon, occupational and ambient exposures.

Evidence and expected outcome. Large causal evidence supports exposure reduction; policy, engineering control and cessation support outperform advice alone.

Risks, burden, and equity. Withdrawal, unequal access to treatment, substitution with unvalidated products, economic and housing constraints.

Response monitoring. Exposure history, biochemical verification when useful, symptoms, lung function, indoor/workplace measures and policy reach.

Evidence anchors: ^[10,36]

12.0.2 Vaccination and infection prevention

Target and mechanism. Reduce respiratory infection, severe disease and transmission through immunization, ventilation, hygiene and risk-specific precautions.

Evidence and expected outcome. Strong but pathogen-, product-, age- and season-specific evidence; benefits include indirect protection.

Risks, burden, and equity. Rare adverse events, waning, mismatch, access, hesitancy and inequitable distribution.

Response monitoring. Coverage, breakthrough severity, adverse-event surveillance, ventilation/air metrics and equity.

Evidence anchors: [14,15]

12.0.3 Inhaled corticosteroid-containing therapy

Target and mechanism. Suppress airway inflammation; central to asthma control and selected COPD phenotypes.

Evidence and expected outcome. Reduces asthma exacerbation risk; benefit in COPD depends on exacerbation and inflammatory phenotype with pneumonia tradeoff.

Risks, burden, and equity. Local candidiasis/dysphonia, systemic exposure at high dose, pneumonia risk in selected COPD, technique and adherence failures.

Response monitoring. Symptoms, exacerbations, reliever use, spirometry, eosinophilic markers where useful, dose and technique.

Evidence anchors: [6,7,38]

12.0.4 Bronchodilators

Target and mechanism. Relax airway smooth muscle through beta-adrenergic or muscarinic pathways.

Evidence and expected outcome. Improve symptoms and function in obstructive disease; regimen depends on diagnosis and severity.

Risks, burden, and equity. Tremor, tachycardia, urinary/ocular effects, paradoxical bronchospasm, overreliance and device mismatch.

Response monitoring. Symptoms, rescue use, heart rate, spirometry, side effects, inhaler technique and adherence.

Evidence anchors: [6,7,37]

12.0.5 Biologic and precision respiratory therapy

Target and mechanism. Block defined immune pathways or correct specific molecular defects such as eligible CFTR variants.

Evidence and expected outcome. Large benefit in selected biomarker/genotype-defined groups; not evidence that all patients with the label share the target.

Risks, burden, and equity. Cost, injection/infusion burden, infection or immune risks, access disparity, nonresponse and uncertain duration.

Response monitoring. Eligibility criteria, exacerbations, function, biomarkers, adverse events, adherence and de-escalation evidence.

Evidence anchors: [7,38,42]

12.0.6 Antimicrobial treatment and stewardship

Target and mechanism. Treat likely or confirmed bacterial, mycobacterial, fungal or viral disease while minimizing resistance and collateral harm.

Evidence and expected outcome. Timely appropriate therapy is lifesaving in severe infection; unnecessary exposure causes harm and resistance.

Risks, burden, and equity. Allergy, toxicity, interactions, microbiome disruption, *Clostridioides difficile*, resistance and diagnostic uncertainty.

Response monitoring. Clinical trajectory, cultures/PCR and susceptibilities, drug levels/organ function when needed, adverse events and local resistance.

Evidence anchors: [12,13]

12.0.7 Oxygen therapy

Target and mechanism. Increase inspired oxygen to correct clinically important hypoxemia.

Evidence and expected outcome. Essential in acute hypoxemia and improves survival in selected chronic severe resting hypoxemia; benefit is not universal for breathlessness without hypoxemia.

Risks, burden, and equity. Hyperoxia, carbon-dioxide retention in susceptible patients, fire risk, dryness, mobility burden and inequitable supply.

Response monitoring. Target saturation by condition, blood gases when ventilation matters, flow/device performance, adherence and functional benefit.

Evidence anchors: [6,27,28]

12.0.8 Noninvasive ventilation and positive airway pressure

Target and mechanism. Apply positive pressure to splint airway, unload muscles, improve ventilation or recruit lung.

Evidence and expected outcome. Established for obstructive sleep apnea and selected acute/chronic ventilatory failure; benefit depends on cause, interface and adherence.

Risks, burden, and equity. Aspiration, skin injury, leaks, intolerance, hemodynamic effects, delayed intubation and algorithm error.

Response monitoring. Gas exchange, work of breathing, synchrony, leak, adherence, pressure data, skin and patient goals.

Evidence anchors: [6,9,27]

12.0.9 Invasive mechanical ventilation

Target and mechanism. Provide controlled or assisted ventilation and oxygenation through an artificial airway.

Evidence and expected outcome. Life-saving when failure cannot be supported otherwise; lung-protective approaches reduce ventilator-induced injury in defined syndromes.

Risks, burden, and equity. Barotrauma/volutrauma, diaphragm weakness, pneumonia, delirium, sedation harm, hemodynamic compromise and long-term disability.

Response monitoring. Pressures/volumes, blood gases, synchrony, sedation, infection, diaphragm and mobility, readiness to liberate.

Evidence anchors: [27,28,29]

12.0.10 Pulmonary rehabilitation

Target and mechanism. Combine exercise training, education, behavior support, breathing/energy strategies and multidisciplinary management.

Evidence and expected outcome. Improves exercise capacity, symptoms and quality of life in chronic respiratory disease and after selected exacerbations.

Risks, burden, and equity. Access, transportation, desaturation risk, comorbidity, inadequate personalization and poor maintenance support.

Response monitoring. Functional tests, symptoms, participation, oxygen response, adherence, goals and adverse events.

Evidence anchors: [6,20,21]

12.0.11 Airway clearance and secretion management

Target and mechanism. Mobilize and remove secretions through hydration, oscillation, positive expiratory pressure, positioning, cough assistance or suction.

Evidence and expected outcome. Established in cystic fibrosis and selected bronchiectasis/neuromuscular phenotypes; technique should match mechanism.

Risks, burden, and equity. Fatigue, reflux/aspiration, desaturation, hemoptysis, device contamination and treatment burden.

Response monitoring. Sputum burden, exacerbations, technique, cough flow, oxygenation and patient-reported feasibility.

Evidence anchors: [16,20]

12.0.12 Surgery, endoscopy and transplantation

Target and mechanism. Resect, repair, drain, reduce volume, reopen airway, remove thrombus or replace failing lungs in selected disease.

Evidence and expected outcome. Can be curative or transformative when anatomy, stage, reserve and goals align.

Risks, burden, and equity. Perioperative death, bleeding, infection, rejection, lifelong immunosuppression, access and opportunity cost.

Response monitoring. Multidisciplinary selection, pulmonary/cardiac reserve, pathology, complications, function, graft status and survival.

Evidence anchors: [17,28]

12.0.13 Palliative and supportive respiratory care

Target and mechanism. Treat breathlessness, cough, anxiety, secretion distress, fatigue and caregiver burden while aligning treatment with goals.

Evidence and expected outcome. Symptom-focused approaches improve experience and can coexist with disease-directed therapy.

Risks, burden, and equity. Sedation, respiratory depression, stigma, fragmented access and late referral.

Response monitoring. Symptoms, function, goals, caregiver burden, adverse effects and crisis plans.

Evidence anchors: [34,35]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can continuous wearable respiratory sensing detect deterioration early enough to improve outcomes?	Wearables can capture rate, motion, oxygenation and activity under selected conditions.	The added clinical utility over symptom plans and intermittent validated testing across diseases and home settings.	Artifact and missingness rise precisely during activity and illness; alert cascades and false reassurance can offset earlier detection.	Pragmatic randomized studies with raw-signal audit, escalation protocols, patient-important outcomes and burden measurement.
How should small-airway disease be defined and monitored?	Pathology can precede abnormal conventional spirometry, and oscillometry, washout and imaging provide complementary proxies.	A transferable threshold or composite that changes treatment and outcomes.	Tests sample different physics and compartments; reference standards and longitudinal intervention evidence are limited.	Prospective multimodal cohorts with bronchoscopy/imaging subsets and treatment-response endpoints.
Can breathomics diagnose disease from exhaled compounds?	Exhaled volatile and aerosol composition carries biological and environmental information.	Reproducible, specific, externally validated signatures that outperform simpler pathways.	Diet, microbiome, ambient contamination, collection materials, humidity and platform drift create shortcuts.	Locked multisite studies with blanks, standardized preanalytics, independent replication and utility trials.
What causes persistent post-infectious respiratory symptoms?	Residual structural disease, vascular dysfunction, autonomic dysregulation, dysfunctional breathing, deconditioning and comorbidities can each contribute.	The prevalence, causal markers and treatment response of mechanistically distinct subgroups.	Broad syndromic definitions and referral bias mix pathways; tests at rest may miss exertional abnormalities.	Deep longitudinal phenotyping with challenge testing, causal intervention trials and functional outcomes.

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Question	What is established	What remains uncertain	Why unresolved	Decisive next step
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within respiratory health and measurement engineering.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

13.0.1 Can continuous wearable respiratory sensing detect deterioration early enough to improve outcomes?

Established knowledge. Wearables can capture rate, motion, oxygenation and activity under selected conditions.

Unresolved issue. The added clinical utility over symptom plans and intermittent validated testing across diseases and home settings.

Why current evidence is insufficient. Artifact and missingness rise precisely during activity and illness; alert cascades and false reassurance can offset earlier detection.

Research that would change confidence. Pragmatic randomized studies with raw-signal audit, escalation protocols, patient-important outcomes and burden measurement.

Evidence anchors: [23,24]

13.0.2 How should small-airway disease be defined and monitored?

Established knowledge. Pathology can precede abnormal conventional spirometry, and oscillometry, washout and imaging provide complementary proxies.

Unresolved issue. A transferable threshold or composite that changes treatment and outcomes.

Why current evidence is insufficient. Tests sample different physics and compartments; reference standards and longitudinal intervention evidence are limited.

Research that would change confidence. Prospective multimodal cohorts with bronchoscopy/imaging subsets and treatment-response endpoints.

Evidence anchors: [6,7,8]

13.0.3 Can breathomics diagnose disease from exhaled compounds?

Established knowledge. Exhaled volatile and aerosol composition carries biological and environmental information.

Unresolved issue. Reproducible, specific, externally validated signatures that outperform simpler pathways.

Why current evidence is insufficient. Diet, microbiome, ambient contamination, collection materials, humidity and platform drift create shortcuts.

Research that would change confidence. Locked multisite studies with blanks, standardized preanalytics, independent replication and utility trials.

Evidence anchors: [24,43]

13.0.4 What causes persistent post-infectious respiratory symptoms?

Established knowledge. Residual structural disease, vascular dysfunction, autonomic dysregulation, dysfunctional breathing, deconditioning and comorbidities can each contribute.

Unresolved issue. The prevalence, causal markers and treatment response of mechanistically distinct subgroups.

Why current evidence is insufficient. Broad syndromic definitions and referral bias mix pathways; tests at rest may miss exertional abnormalities.

Research that would change confidence. Deep longitudinal phenotyping with challenge testing, causal intervention trials and functional outcomes.

Evidence anchors: ^[20]

13.0.5 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within respiratory health and measurement engineering.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Evidence anchors: ^[42]

13.0.6 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: ^[24]

13.0.7 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[17]

13.0.8 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[23]

13.0.9 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: ^[21]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Respiratory digital twin falsification	Can a compact model predict gas exchange and work of breathing across posture, activity and disease?	Simultaneous reference airflow, pressures, gases, oximetry, motion and imaging in a diverse challenge cohort; preregister model structure and failure states.	Individual prediction intervals remain calibrated during held-out disease states and interventions.	Model fits average waveforms but fails during clinically important transitions or requires unobservable parameters.
0. Equity-critical oximetry study	What conditions produce clinically consequential saturation error across skin characteristics, perfusion and disease?	Prospective paired co-oximetry and multiple devices across full range; model repeated measures, device lot, motion and perfusion.	Prespecified error and occult-hypoxemia limits met or actionable correction/abstention rules validated.	Average accuracy hides subgroup or low-perfusion failures.
0. Home exacerbation utility trial	Does a synchronized low-burden home stack reduce severe asthma/COPD exacerbations without alarm harm?	Pragmatic randomized trial comparing current action plans with device-assisted plans; measure admissions, treatment, false alerts, adherence and equity.	Net reduction in severe events with acceptable burden and no subgroup harm.	More alerts and treatment with no patient-important benefit.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for respiratory health and measurement engineering?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.

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Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

14.0.1 0. Respiratory digital twin falsification

Question. Can a compact model predict gas exchange and work of breathing across posture, activity and disease?

Design. Simultaneous reference airflow, pressures, gases, oximetry, motion and imaging in a diverse challenge cohort; preregister model structure and failure states.

Success criterion. Individual prediction intervals remain calibrated during held-out disease states and interventions.

Failure or stop criterion. Model fits average waveforms but fails during clinically important transitions or requires unobservable parameters.

Evidence anchors: [8,24]

14.0.2 0. Equity-critical oximetry study

Question. What conditions produce clinically consequential saturation error across skin characteristics, perfusion and disease?

Design. Prospective paired co-oximetry and multiple devices across full range; model repeated measures, device lot, motion and perfusion.

Success criterion. Prespecified error and occult-hypoxemia limits met or actionable correction/abstention rules validated.

Failure or stop criterion. Average accuracy hides subgroup or low-perfusion failures.

Evidence anchors: [22,23]

14.0.3 0. Home exacerbation utility trial

Question. Does a synchronized low-burden home stack reduce severe asthma/COPD exacerbations without alarm harm?

Design. Pragmatic randomized trial comparing current action plans with device-assisted plans; measure admissions, treatment, false alerts, adherence and equity.

Success criterion. Net reduction in severe events with acceptable burden and no subgroup harm.

Failure or stop criterion. More alerts and treatment with no patient-important benefit.

Evidence anchors: [6,7,45]

14.0.4 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for respiratory health and measurement engineering?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

Evidence anchors: [22,43]

14.0.5 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

14.0.6 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: [24,26]

14.0.7 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: [45]

14.0.8 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

Evidence anchors: [21]

14.0.9 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: ^[22,45]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03A	Cardiovascular Health	Cardiopulmonary coupling, oxygen delivery, pulmonary circulation, embolism and exercise limitation.	interpreting breathlessness, desaturation, pulmonary hypertension or exercise tests.
03C	Nervous System Health	Control of breathing, neuromuscular pump, cough, swallowing and autonomic regulation.	ventilation or airway protection fails without primary parenchymal disease.
03G	Hematologic Health	Hemoglobin, thrombosis, dyshemoglobinemia and oxygen-carrying capacity.	oxygen saturation, embolism or dyspnea does not match lung findings.
08	Infection, Immunity, Host-Microbe and Biosecurity	Respiratory pathogens, vaccination, AMR, microbiome and immune response.	infection, recurrent exacerbation or stewardship is central.
09	Environmental and Occupational Health	Air pollution, indoor air, workplace hazards, climate and exposure measurement.	prevention or causal attribution depends on exposure.
10	Function, Disability and Rehabilitation	Breathlessness, activity, participation, pulmonary rehabilitation and assistive technology.	clinical values must be translated into daily capability.
12	Health Care and Delivery Systems	Emergency, intensive, home oxygen/ventilation and longitudinal care pathways.	implementation, access or escalation is part of the outcome.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Alveolar ventilation	The portion of minute ventilation reaching gas-exchanging units after dead-space ventilation is excluded.
Compliance	Change in volume per change in distending pressure; a property of the coupled lung-chest system under stated conditions.
Dead space	Ventilated volume that does not participate effectively in carbon-dioxide exchange, including anatomic and physiologic components.
Hypoxemia	Low oxygen tension or saturation in arterial blood; distinct from tissue hypoxia and low oxygen delivery.
Shunt	Perfusion of blood through regions that do not receive effective ventilation, limiting correction by supplemental oxygen depending on magnitude.
Ventilation-perfusion matching	Regional coordination of air delivery and blood flow required for efficient gas exchange.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.